A disk-lens counterexample to Lipschitz regularity under infimal convolution

Candidate solution packet for arXiv:1710.08233

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Status. Candidate counterexample, likely valid, pending human review.

Source. Simon Zugmeyer, *Sharp trace Gagliardo–Nirenberg–Sobolev inequalities for convex cones, and convex domains*, arXiv:1710.08233 [1], Section 2.2, source PDF page 10. For extended-real functions, the source uses

$$(f \,\square\, g)(x) = \inf_{y \in \mathbb{R}^n} \{f(y) + g(x-y)\}, \qquad \operatorname{dom} f = \{x : f(x) < +\infty\}.$$

After a nonconvex, empty-interior example, it conjectures that Lipschitz functions on convex domain(s) with nonempty interior have a Lipschitz infimal convolution.

then

$$(f \,\square\, g)(x_1, x_2) = \begin{cases} 1 & \text{if } x_1 \in (0,1],\ x_2 \in [0,1], \\ 1 - x_2 & \text{if } x_1 = 0,\ x_2 \in [0,1], \\ 0 & \text{if } x_1 = 0,\ x_2 \in [1,2], \\ +\infty & \text{otherwise} \end{cases}$$

is not a continuous function. This example can easily be adapted to obtain a discontinuous infimal convolution for smooth functions f and g. We conjecture that if the domain is assumed convex, and if both functions are Lipschitz continuous, and their domain is of non-empty interior, then their infimal convolution is Lipschitz continuous.

Lemma 2.8, together with Lemma 2.6 and Rademacher's theorem, prove the following proposition:

Proof intuition

For two unit disks, the feasible decompositions of a point x form the lens $D \cap (x - D)$. Near the boundary of $D + D$, this lens closes tangentially: its vertical radius is proportional to the square root of the distance to the boundary. An affine function that reads the vertical coordinate makes the infimal convolution select the bottom of the lens. The resulting value function therefore has a square-root cusp, even though both input functions are Lipschitz on the same smooth compact convex domain.

Counterexample

Let

$$D = \{(u_1, u_2) \in \mathbb{R}^2 : u_1^2 + u_2^2 \le 1\}$$

and define $f, g : \mathbb{R}^2 \to \mathbb{R} \cup \{+\infty\}$ by

$$f(u_1, u_2) = \begin{cases} 1 + u_2, & (u_1, u_2) \in D, \\ +\infty, & \text{otherwise,} \end{cases} \qquad g(u_1, u_2) = \begin{cases} 0, & (u_1, u_2) \in D, \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 1. *The functions f and g have the same compact convex domain D, whose interior is nonempty. Their restrictions to D are respectively 1-Lipschitz and 0-Lipschitz. Both extended-real functions are proper, lower semicontinuous, convex, and nonnegative, and $f \,\square\, g$ is exact. Nevertheless, $f \,\square\, g$ is not Lipschitz on its domain $D + D = 2D$; in fact it is not locally Lipschitz at $(2, 0)$ relative to $2D$.*

Proof. All input claims are immediate: f is an affine function plus the convex indicator of the closed disk, while g is that indicator. Compactness of the feasible set gives exactness at every point of $2D$.

For $0 \le t \le 2$, write $x_t = (t, 0)$. A decomposition $x_t = a + b$ with $a, b \in D$ is equivalent, on writing $a = (s, r)$, to

$$s^2 + r^2 \le 1, \qquad (t - s)^2 + r^2 \le 1. \tag{1}$$

Thus

$$\begin{aligned}(f \,\square\, g)(x_t) &= 1 + \min\{r : (s, r) \text{ satisfies (1)}\}\\ &= 1 - \sqrt{1 - \min_{s\in\mathbb{R}} \max\{s^2, (t - s)^2\}}.\end{aligned}$$

The scalar minimax is exact. Indeed,

$$t \le |s| + |t - s| \le 2 \max\{|s|, |t - s|\},$$

so $\max\{s^2, (t - s)^2\} \ge t^2/4$, with equality for $s = t/2$. Consequently

$$(f \,\square\, g)(x_t) = 1 - \sqrt{1 - \frac{t^2}{4}} \qquad (0 \le t \le 2). \tag{2}$$

For $0 < \varepsilon < 2$, formula (2) gives

$$\frac{|(f \,\square\, g)(x_2) - (f \,\square\, g)(x_{2-\varepsilon})|}{\|x_2 - x_{2-\varepsilon}\|_2} = \frac{\sqrt{\varepsilon - \varepsilon^2/4}}{\varepsilon} = \sqrt{\frac{1}{\varepsilon} - \frac{1}{4}} \longrightarrow +\infty.$$

No finite Lipschitz constant exists, even in any relative neighborhood of $(2, 0)$ in $2D$. □

Wording variants and scope

Remark 1 (Open domains)**.** *If "domain" was intended to mean an open convex set, replace D by its open unit disk. Formula (2) holds for $0 \le t < 2$. Both $x_{2-\varepsilon}$ and $x_{2-2\varepsilon}$ are interior points, while*

$$\frac{|(f \,\square\, g)(x_{2-\varepsilon}) - (f \,\square\, g)(x_{2-2\varepsilon})|}{\varepsilon} = \frac{\sqrt{2\varepsilon - \varepsilon^2} - \sqrt{\varepsilon - \varepsilon^2/4}}{\varepsilon} \sim \frac{\sqrt{2} - 1}{\sqrt{\varepsilon}}.$$

Hence the convolution is still not globally Lipschitz on its open domain.

The example refutes the printed global-Lipschitz conjecture. It does not refute the weaker assertion that the convolution is locally Lipschitz at each point of $\mathrm{int}(D + D)$; the closed-disk failure occurs at the boundary, and the open-disk function is locally Lipschitz at each fixed point on the displayed diameter. If a factor is finite and globally L-Lipschitz on all of $\mathbb{R}^n$, then its infimal convolution with any function is L-Lipschitz whenever finite, so the extended-domain interpretation is the only nontrivial one in the source context.

Verification and novelty

The proof is exact and uses no computation. The bundled checker tests the scalar minimax inequality with rational arithmetic on 151,151 grid pairs and prints the divergent quotients for closed and open disks. All tests pass.

A bounded search on 17 August 2026 covered the four lightweight run indexes, the exact arXiv id and title, the exact conjecture sentence, the author name, and close arXiv/web queries combining "infimal convolution", "convex domain", "nonempty interior", and "Lipschitz". The exact-phrase hit was the published version of the source itself. No later answer or matching disk counterexample was found. Novelty confidence is moderate: an elementary example may exist as folklore or under different terminology.

Human-review recommendation. Check that the source's "Lipschitz continuous" is read globally on the essential domain, as its wording and surrounding extended-real definitions indicate. The formal construction and lens computation are otherwise elementary.

References

[1] S. Zugmeyer, *Sharp trace Gagliardo–Nirenberg–Sobolev inequalities for convex cones, and convex domains*, Ann. Inst. H. Poincaré Anal. Non Linéaire 36 (2019), no. 3, 861–885; arXiv:1710.08233.